

Logarithmic scalar-field profiles near the Schwarzschild singularity

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We ask whether a nonlinear vacuum potential leaves an imprint on scalar-field profiles near the Schwarzschild singularity while the spacetime metric is held fixed. We extract the Fournodavlos–Sbierski logarithmic coefficient from paired, identical-data finite-element evolutions with excision and compare a Mexican-hat scalar mode by mode with a free scalar. The production Mexican-hat-to-linear L^2 ratios are 0.923, 0.867, and 0.876 for $l = 0, 1, 2$, with a conservative 10–15% numerical budget. Their order-unity values exclude a parametrically suppressed nonlinear coefficient at the measured point. A profile-dependent window calibration explains 31.0% of the monopole L^2 deficit on common support and leaves 4.16 percentage points as a three-dimensional mesh/extraction systematic. Exterior validation gives $M\omega = 0.48772 \pm 0.01244 - i(0.10161 \pm 0.00545)$ from late, large-domain windows. The result is restricted to the quantified Cowling, or test-field, regime and does not include metric backreaction.

I. INTRODUCTION

For a massless scalar on a fixed Schwarzschild spacetime, Fournodavlos and Sbierski proved that smooth solutions approaching the spacelike singularity admit the expansion

$$\phi(t, r, \Omega) = A(t, \Omega) \ln r + B(t, \Omega) + \mathcal{O}(r |\ln r|), \quad (1)$$

and that logarithmic blow-up is generic in precise energy topologies [1]. The coefficient A is a function on the singular hypersurface rather than a single global amplitude. Its spherical harmonic components are the observables $a_l(t)$ studied below.

The theorem is linear, while event-horizon data are themselves controlled by precise Price-law asymptotics [2]. The approach to a spacelike singularity is often associated with velocity or kinetic domination. In self-gravitating Einstein–scalar systems, monotone, quiescent singular behavior is stable in broad near-FLRW and subcritical Kasner regimes, with increasingly general formation and complete-asymptotic results [3–6]. Perturbed near-Schwarzschild interiors likewise retain a spacelike singular regime, although backreaction changes its detailed geometry [7, 8]. This motivates a test-field question with a direct numerical answer: does a nonlinear vacuum potential retain an observable imprint in $A(t, \Omega)$, or is that structure washed out as the kinetic terms dominate?

Previous numerical studies provide complementary pieces but not this measurement. Self-gravitating spherical calculations established an early numerical lineage for deep black-hole interiors [9]. Thuestad, Khanna, and Price evolved test scalar fields through the interiors of Schwarzschild and Kerr black holes and found interior l -dependent oscillations, without extracting the $r \rightarrow 0$ logarithmic coefficient [10]. Symmetry-breaking scalars have also been evolved in the exterior: accretion can delay symmetry breaking near the horizon [11] or form symmetry-restored shells and trigger vacuum decay [12]. Black-hole-seeded vacuum decay itself has a substantial literature, with both catalytic scenarios and strong counterarguments [13, 14]. These works motivate the physics

while leaving the near-singularity profile open.

We present, to our knowledge, the first numerical measurement of $A(t, \Omega)$ approaching the Schwarzschild singularity and the first identical-data comparison of its free and symmetry-breaking evolutions. The measurement is performed mode by mode in $3 + 1$ dimensions. The methodology is deliberately not claimed as the first scalar evolution with excision: three-dimensional finite-difference excision predates this work [15], and finite elements have been used for black-hole perturbations [16]. The methodological contribution is the combination of a DOLFINx finite-element evolution, calibrated multi-radius interior extraction, paired potential comparison, and an exterior validation that separates finite-domain and overtone systematics.

Section II defines the physical problem and its scope. Section III describes the evolution and observables. Sections IV and V give the interior and exterior results, followed by limitations and interpretation in Sec. VI.

II. PHYSICAL SETUP AND SCOPE

We evolve a real scalar on a Schwarzschild background in ingoing Kerr–Schild coordinates. With signature $(-, +, +, +)$,

$$g_{\mu\nu} = \eta_{\mu\nu} + 2H k_\mu k_\nu, \quad H = \frac{M}{r}, \quad k_\mu = (1, x_i/r), \quad (2)$$

which is regular at the event horizon. Geometrized units $G = c = M = 1$ are used unless M is shown explicitly.

The scalar action and equation of motion are

$$S_\phi = - \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right], \quad (3)$$

$$\square_g \phi - V'(\phi) = 0. \quad (4)$$

Two models are compared. The linear member has $V_{\text{lin}} = 0$. The symmetry-breaking member has

$$V_{\text{hat}}(\phi) = \frac{\lambda}{4} (\phi^2 - v^2)^2. \quad (5)$$

We use $\lambda = 0.10$ and $v = 1.0$, and initialize the nonlinear field in the vacuum approached at the outer boundary. The two members of every comparison receive the same perturbation.

The production perturbation is an ingoing curved Gaussian with amplitude 0.10, center $r_0 = 5.0M$, and width $w = 1.0M$, multiplied by a real Y_{l0} . We evolve $l = 0, 1, 2$. The angular decomposition is performed numerically in three dimensions; no spherical symmetry is imposed on the evolution.

The metric is not evolved. This Cowling or test-field limit is a physical restriction of the study, not an omitted convergence check. In the one-dimensional pilot at the largest production amplitude, 0.10, the dimensionless backreaction monitor $\zeta = 8\pi\rho/\sqrt{K_{\mu\nu\rho\sigma}K^{\mu\nu\rho\sigma}}$ stays below 0.00124 inside the horizon and falls to 0.00016 at the deepest sampled radius. The global three-dimensional monitor is instead of order unity to ten because the weak-curvature exterior dominates its denominator; it is not an interior Cowling measure. The fixed-background scope matches Eq. (1) but does not address backreaction, mass inflation, or singularity resolution.

III. NUMERICAL METHOD AND OBSERVABLES

A. First-order finite-element evolution

Writing the metric in 3+1 form with lapse α , shift β^i , spatial metric γ_{ij} , and extrinsic-curvature trace K , we define

$$\Pi = \alpha^{-1}(\partial_t\phi - \beta^i\partial_i\phi). \quad (6)$$

The evolved system is

$$\partial_t\phi = \alpha\Pi + \beta^i\partial_i\phi, \quad (7)$$

$$\partial_t\Pi = \alpha D_i D^i\phi + D^i\alpha D_i\phi + \beta^i\partial_i\Pi + \alpha K\Pi - \alpha V'(\phi). \quad (8)$$

We use the sign convention $K_{ij} = -\frac{1}{2}\mathcal{L}_n\gamma_{ij}$, for which the momentum equation contains the displayed $+\alpha K\Pi$ term. We discretize both variables with continuous piecewise-linear elements on a graded tetrahedral Gmsh mesh [17] and assemble the weak form in DOLFINx [18]. PETSc and its Python bindings supply the distributed linear algebra [19, 20]. Time integration uses SSP-RK3 [21]. The black-hole interior is excised inside the horizon, where no incoming characteristic requires inner boundary data. At the outer boundary we use a characteristic radiative condition; exterior spectroscopy additionally uses a smooth damping layer. Because absorbing-boundary performance is curvature- and multipole-dependent [22], we quantify the finite-domain response by a matched-domain experiment rather than assuming it negligible.

In the causally clean transit-and-ringdown interval, waveform self-convergence is approximately second order

on the coarse triplet and remains convergent, with reduced observed order, on the fine triplet. The residual reduction is consistent with non-nested Gmsh remeshing noise and is unchanged by replacing faceted with curved cells; the later cavity-dominated signal is excluded from convergence claims.

We record the Killing-energy change together with the inner and outer fluxes as an implementation diagnostic. Its production residual is dominated by surface quadrature on the faceted excision boundary, so it is not promoted as a field-accuracy metric; convergence of the evolved field and the paired profile observables is assessed separately.

The linear operator is preassembled. Only the cubic part of $V'_{\text{hat}} = \lambda\phi(\phi^2 - v^2)$ is assembled at each stage. The consistent mass matrix is retained in production. A direct A/B test found that row-sum mass lumping displaced the interior observable by 27.7% in the median and 38.6% at maximum. It was therefore rejected despite its computational speed.

B. Near-singularity estimator

Following the full hierarchy associated with Eq. (1), the primary radial model for each multipole is

$$\phi_l(t, r) \simeq a_l(t) \ln r + b_l(t) + c_l(t)r \ln r + d_l(t)r. \quad (9)$$

The fit is performed over $r/M \in [0.10, 0.50]$ with truncation order 1. A lower-order fit over the same radii supplies only the phase anchor. The primary order is a bias-variance compromise: higher orders reduce the one-dimensional truncation bias but amplify correlated three-dimensional mesh errors.

Statistics use the anchored strong phase $t/M \in [4.0, 10.0]$ and require the anchor magnitude to exceed 0.30 of its maximum in that interval. This upper time cut removes late finite-domain contamination. We define the deep-window one-dimensional reference by applying the same asymptotic order on $r/M \in [0.02, 0.20]$ at order 1.

The kinetic mechanism is checked independently in the production-protocol Mexican-hat oracle using the dynamical ratio $R_V = |\alpha V'(\phi)|$ divided by the sum of the absolute non-potential terms in $\partial_t\Pi$. Across strong-phase snapshots, the contiguous region with $R_V < 0.01$ reaches a median $r_* = 0.625M$ [IQR 0.414M, 0.808M], while the deep radial scaling is $R_V \propto r^{2.79}$ [IQR 2.44, 3.29]. Thus the deep-window reference lies entirely within the kinetically dominated region, while the primary interval samples both its interior and the time-dependent crossover.

C. Paired discriminator and calibration

The primary identical-data discriminator for each l is

$$\mathcal{D}_l = \frac{\|a_{\text{hat},l}\|_{2,\mathcal{S}}}{\|a_{\text{lin},l}\|_{2,\mathcal{S}}}, \quad (10)$$

where $\mathcal{S}$ is the common strong support. The median and interquartile range of $a_{\text{hat},l}/a_{\text{lin},l}$ are secondary summaries. Pointwise ratios near a zero crossing are ill-conditioned. Ratios of separate maxima are excluded because the maxima occur at different times and inherit a differential window bias.

For $l = 0$, a diagnostic calibration is obtained by applying the production fit and the deep-window reference fit to the same dense oracle snapshot,

$$c_X(t) = \frac{a_{X,\text{prod}}(t)}{a_{X,\text{ref}}(t)}. \quad (11)$$

The ratio is defined only in the oracle strong phase and is never filled across its zero-crossing gap or beyond its support. Corrections are applied to both members of a pair before recomputing a diagnostic discriminator. They do not replace the production values (frozen before the calibration analysis).

IV. INTERIOR RESULTS

A. Logarithmic profiles

Figure 1 compares the three-dimensional monopole profiles with the dense spherical oracle and Eq. (9). The curves are approximately affine in $\ln r$, and the same basis describes both potentials across representative parts of the common strong phase.

At the finest production resolution level, the median three-dimensional-to-oracle deviation is 13.1% for the linear field and 8.6% for the Mexican-hat field. These comparisons include temporal interpolation and extraction systematics; they are not pure mesh uncertainties.

B. Identical-data discriminator

The mode-by-mode result is summarized in Table I and Fig. 2. In every mode the nonlinear coefficient remains of the same order as its linear partner. This is the operational signature of vacuum-structure erasure: the potential changes amplitudes within the numerical budget but does not introduce a distinct near-singularity scaling.

For $l = 1$ and $l = 2$, the reference-window floors are 10.1% and 6.6%, respectively. They are comparable to or larger than the effect one would attempt to correct, so promoting either an intrinsic or transferred $l > 0$ correction would add unjustified precision. The fine-resolution RMS differences for the individual monopole series are 9.3% and 11.8% for the linear and nonlinear runs. Their paired discriminator is visibly more stable than either absolute comparison because the members share the mesh, estimator, and initial data. Across all three monopole resolution levels, the primary paired discriminator varies by only 2.1 percentage points.

If the vacuum structure survived dynamically, the nonlinear member would be expected to remain anchored

TABLE I. Production L^2 discriminator at the finest available resolution level. The $l > 0$ values are not calibration-corrected; their reference-window floors are stated in the text. Table note: the broad IQR reflects the ill-conditioning of the pointwise ratio near a zero crossing; $\mathcal{D}_l$ is the primary statistic.

l	$\mathcal{D}_l$	median [IQR]
0	0.923	0.851 [0.574, 1.175]
1	0.867	0.640 [0.329, 0.884]
2	0.876	0.813 [0.488, 0.975]

near its vacuum and to yield $\mathcal{D}_l \ll 1$, parametrically separated from an order-unity response. The observed range $\mathcal{D}_l = 0.867\text{--}0.923$, together with the conservative 10–15% budget, excludes that suppression at the measured point. This statement initially covers one potential shape, one (λ, v) pair, and one perturbation amplitude; the one-dimensional sensitivity scan in Sec. VI tests the nearby parameter range.

C. Profile-dependent window systematic

On their oracle strong phases, the sampled calibration factors have medians 1.020 [0.982, 1.042] and 1.014 [0.993, 1.069] for the linear and Mexican-hat profiles. Figure 3 displays their time dependence and the strict-support deviation budget.

For the fine linear run the common-support median deviation changes from 10.31% to 7.11%, accounting for 31.0% of that median deviation. The corresponding Mexican-hat values are 4.83%, 3.89%, and 19.3%. These are systematic diagnostics on reduced support, not substitutes for the production values quoted above.

D. Off-channel response

Nonlinear mode coupling is constrained by comparing each off-channel signal with its linear control over the same capped strong phase. For the fine $l = 2$ pair, the $l = 0$ off-channel fractions are 1.95% (linear) and 1.89% (nonlinear), while the $l = 4$ fractions are 3.38% and 3.40%. The pairs are indistinguishable at this level. The $l = 1 \rightarrow 0$ pair changes from 1.61% to 3.50% and is the only weak excess. Only an excess over the paired linear junk constrains coupling; no isolated fraction is treated as a detection.

V. EXTERIOR VALIDATION

We validate the same three-dimensional evolution and extraction stack using the fundamental scalar

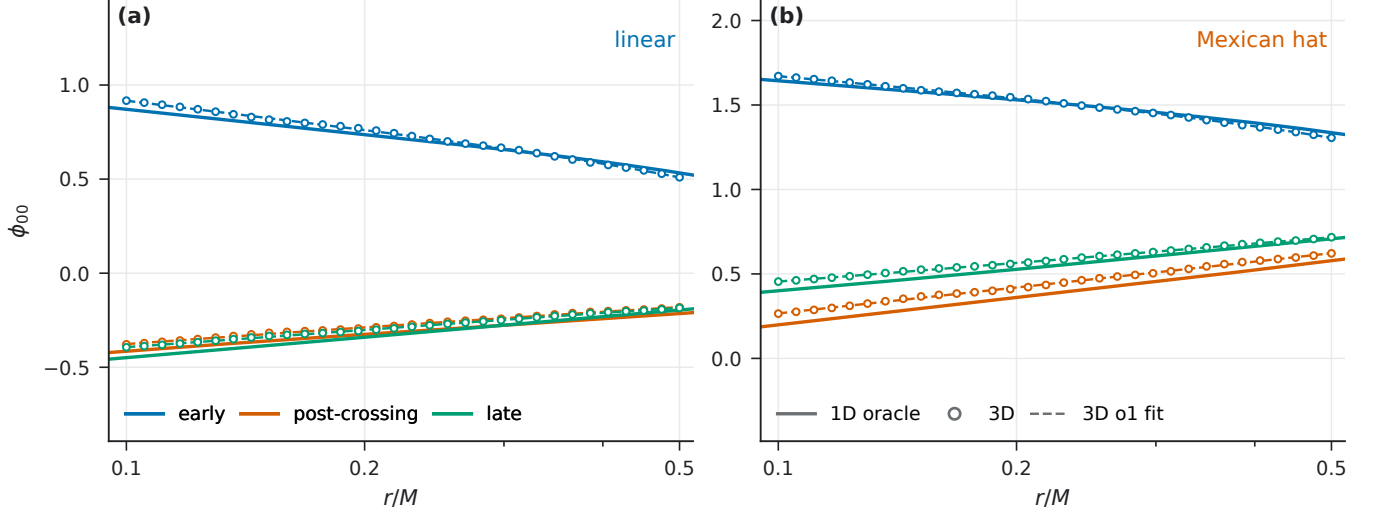


FIG. 1. Near-singularity monopole profiles for identical initial data in the linear and Mexican-hat models. Solid curves show the dense spherical oracle, open circles the versioned mid-resolution production level ($\ell_c = 0.040M$), and dashed curves the primary order-one fits. Here $\phi_{00} = c_{00}/\sqrt{4\pi}$ is the Y_{00} component in the spherical normalization. Colors select reproducible representatives of the common strong phase. This is a qualitative profile comparison; quantitative deviations and the L^2 discriminator are reported separately.

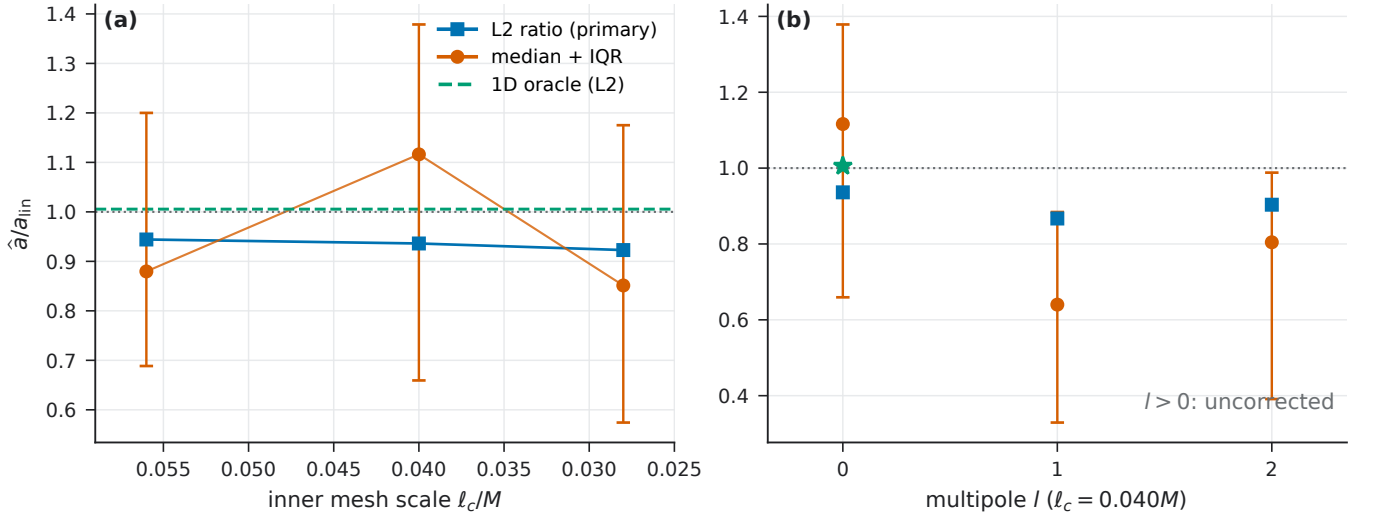


FIG. 2. Identical-data Mexican-hat-to-linear discriminator. Squares show the production L^2 ratio, the primary statistic; circles and bars show the secondary median and IQR. The dashed line is the spherical-oracle L^2 ratio. The $l > 0$ values are uncorrected because their reference-window floors exceed the correction being tested. Peak ratios and corrected common-support L^2 diagnostics are deliberately omitted.

$l = 2$ Schwarzschild quasinormal mode. The reference frequency is obtained independently by the Leaver continued-fraction method [23]. Its value is $M\omega_L = 0.48364 - i0.09676$.

We excite $(l, m) = (2, 0)$ and extract at $r = 6.0M$. The exterior pulse has amplitude 0.001, center $8.0M$, and width $2.0M$. The designed fit uses peak-anchored windows of length $26.0M$, begins its peak search at $12.0M$, and permits 4 Prony modes.

A. Real frequency and damping

Figure 4 separates two effects that a single window average would mix: mesh behavior of the real frequency and first-overtone bias of the damping rate.

At the fine small-domain resolution level, $\text{Re}(M\omega) = 0.47455$ with signed error -1.88% relative to Leaver. The fine-pair order is $p = 1.77$; a three-point Richardson extrapolation is not used because the coarse resolution level lies outside a monotone asymptotic regime.

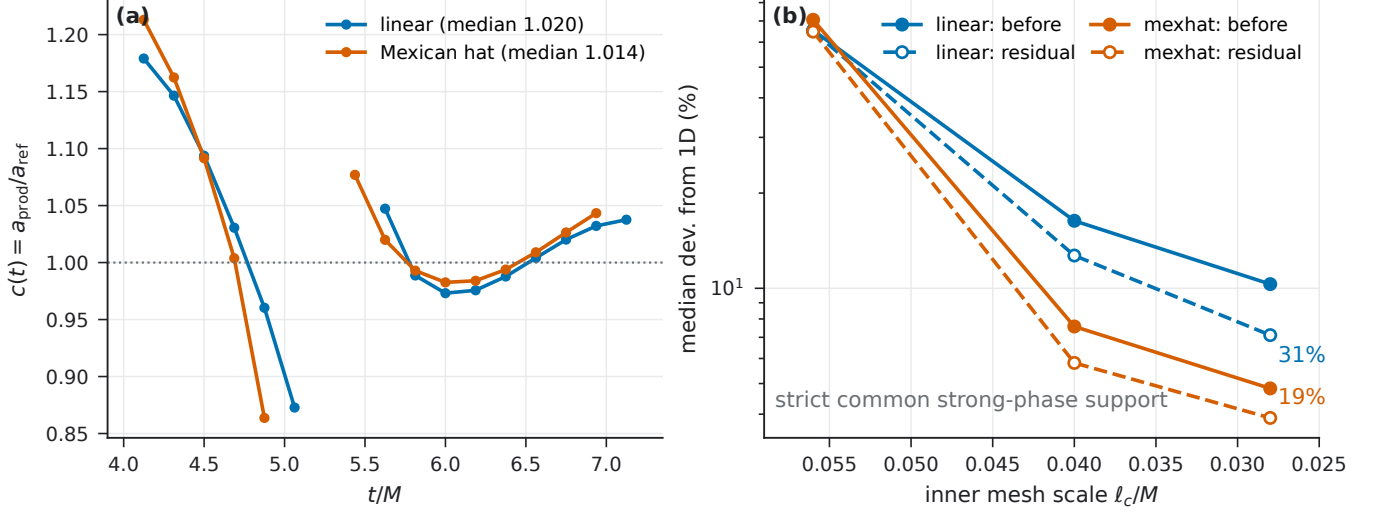


FIG. 3. Window calibration of the monopole estimator. Left: $c(t)$ for the linear and Mexican-hat oracles, defined only on strong-phase support and left unfilled elsewhere. Right: median 3D–1D deviation before and after the diagnostic calibration on strict common support. Residual curves do not replace the production deviations; the coarsest resolution level is shown diagnostically and is not used for a convergence claim.

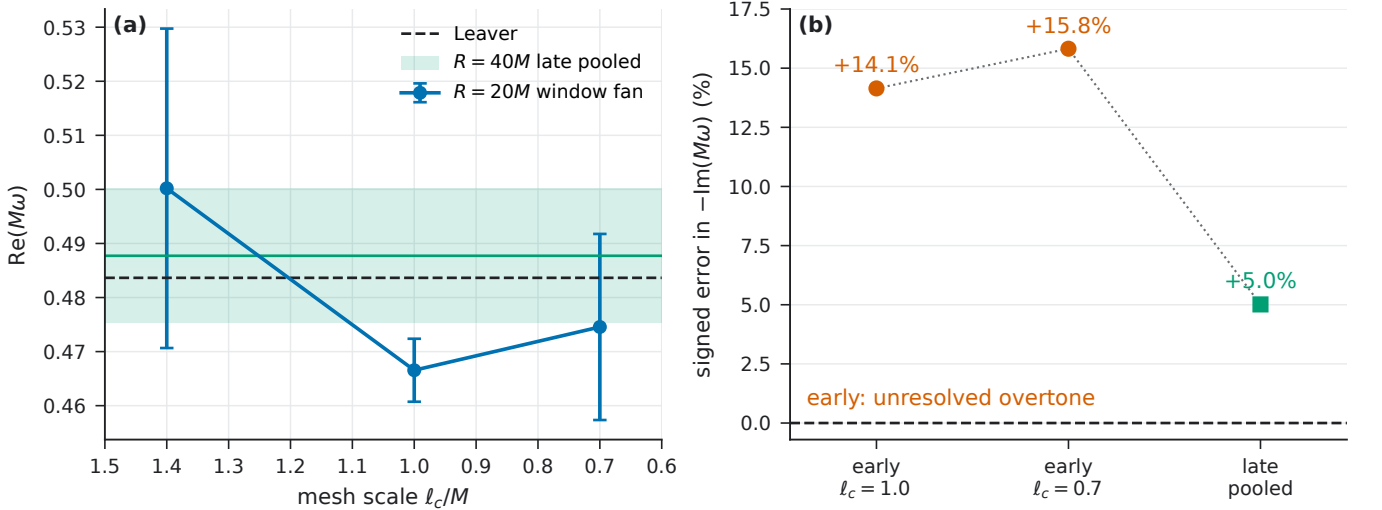


FIG. 4. Exterior $l = 2$ spectroscopy. Left: real-frequency ladder from anchored window fans, with the Leaver value and the late pooled large-domain band. Error bars and the band represent fit/window scatter, not pure discretization uncertainty. Right: early-window damping biases from the unresolved first overtone compared with the late pooled estimate. The early points are protocol biases, not fundamental-mode measurements; the retracted small-domain damping estimate is not shown.

Pooling declared late windows from two matched large-domain resolution levels gives

$$\text{Re}(M\omega) = 0.48772 \pm 0.01244, \quad (12)$$

$$-\text{Im}(M\omega) = 0.10161 \pm 0.00545. \quad (13)$$

The signed errors are +0.84% and +5.01%. These uncertainties are scatter across late windows and two resolution levels and are dominated by the fitting protocol. Early large-domain windows instead overestimate the damping by 14.1% and 15.8%. The first overtone is not separable in the adopted finite window, so those val-

ues quantify a protocol bias rather than alternative measurements of the fundamental. This controlled scalar example mirrors the broader ringdown start-time and overtone debate [24, 25].

B. Finite-domain cavity

The small exterior domain develops a late trapped signal between the black-hole potential barrier and the damping layer. A matched large domain preserves the

inner mesh grading while moving the layer outward, isolating this domain effect.

Figure 5 shows the paired waveform and tail spectrum. The matched enlargement reduces the tail floor by a factor 47.1 and raises peak-to-floor from 4.1 to 195.5. The small-domain lines are $M\omega_1 = 0.3535 \pm 0.0010$ and $M\omega_2 = 0.5630 \pm 0.0040$. They are trapped-domain diagnostics, not physical quasinormal frequencies. The matched-domain real-frequency shift, $\Delta\text{Re}(M\omega) = -0.00221 \pm 0.00582$, remains within the window scatter.

VI. DISCUSSION

The central result is not equality of the two potentials to arbitrary precision. It is the persistence of the same logarithmic asymptotic coefficient at order unity for identical data, in all measured multipoles, with a separately measured error budget. Absolute three-dimensional-to-one-dimensional deviations are larger than the resolution-level spread of the paired discriminator because correlated mesh and estimator errors cancel to first order in the pair.

The monopole calibration classifies the window bias as a partial explanation: it accounts for 31.0% of the L^2 deficit on common support and leaves 4.16 percentage points. The residual is therefore carried as a genuine three-dimensional mesh/extraction systematic. We do not use the corrected diagnostic as a replacement result, and we do not transfer it to $l > 0$.

To test the restriction to one production point, we performed a 12-cell, $l = 0$, $v = 1$ oracle scan over $\lambda \in [0.03, 1.00]$ and $A \in [0.01, 0.10]$, with the same strong-phase discriminator as in production. It gives $\mathcal{D}_0^{\text{oracle}} = 0.712\text{--}1.313$ and a maximum distance from unity of 0.313, above the predeclared review threshold 0.15. At every sampled amplitude, the $\lambda = 0.30$ cells lie below unity whereas the $\lambda = 1.00$ cells lie above it. The near-unity result at the production coupling therefore identifies a finite coupling regime, not uniform robustness across the full scan; within the tested grid the coupling dependence dominates the amplitude dependence. This one-dimensional transient boundary neither changes the production three-dimensional value nor enters its mesh/extraction uncertainty budget.

This behavior is complementary to exterior symmetry-restoration studies [11, 12]. Near the horizon, redshift and accretion can delay or reorganize vacuum dynamics; deeper inside, our fixed-background calculation shows that the coefficient controlling the singular profile is instead dominated by the common kinetic evolution. Modified-gravity scenarios can produce qualitatively different conclusions. For example, Bars finds Higgs-controlled extensions through the singularity in a locally conformal framework with antigravity sectors [26]. That contrast does not compete with the present result: our claim is explicitly restricted to a minimally coupled scalar on a fixed GR background.

The main limitations are accordingly clear. Backreaction is absent; the calculation does not decide the geometry of a self-gravitating singularity. The radial fit covers a finite interval rather than the mathematical limit. The $l > 0$ reference-window floors prevent a citable profile correction. The interior strong phase ends before the late quasi-stationary regime, and the exterior damping uncertainty remains protocol dominated. In particular, Ref. [10] predicts late-time spatial nodes proportional to $P_l(r/M - 1)$ across the interior, rather than short-time temporal oscillations at a few extraction radii. Our versioned production bank ends at $12.0M$ and samples only the finite fitting interval; it therefore cannot test that asymptotic spatial statement. A genuine cross-check requires later-time full-interior profiles and is deferred.

VII. CONCLUSIONS

We have extracted the Fournodavlos–Sbierski logarithmic coefficient from three-dimensional finite-element evolutions inside Schwarzschild and compared a free scalar with a symmetry-breaking scalar using identical data. The production primary ratios 0.923, 0.867, and 0.876 show that the nonlinear vacuum structure does not generate a parametrically different near-singularity profile in $l = 0, 1, 2$. A conservative calibration isolates part of the monopole window bias without promoting a correction beyond its support. Independent exterior spectroscopy reproduces the Schwarzschild QNM within a protocol-dominated uncertainty and exposes both the unresolved-overtone bias and a finite-domain cavity floor. Together, these results support kinetic-domination erasure of vacuum structure in the test-field Schwarzschild interior while making the Cowling limit and all resolved systematics explicit.

VIII. DATA AVAILABILITY

Versioned artifacts and scripts regenerate every numerical entry, figure, and the manuscript PDF. A SHA-256 source manifest closes the release inventory. At deposit, the archival address will replace the token below.

[REPOSITORY-URL-AT-DEPOSIT]

Appendix A: Initial data

This appendix fixes the initial-data convention sufficiently to reproduce it independently. Define the radial perturbation and its derivative by

$$g(r) = A \exp\left[-\frac{(r - r_0)^2}{w^2}\right], \quad g'(r) = -\frac{2(r - r_0)}{w^2} g(r). \quad (\text{A1})$$

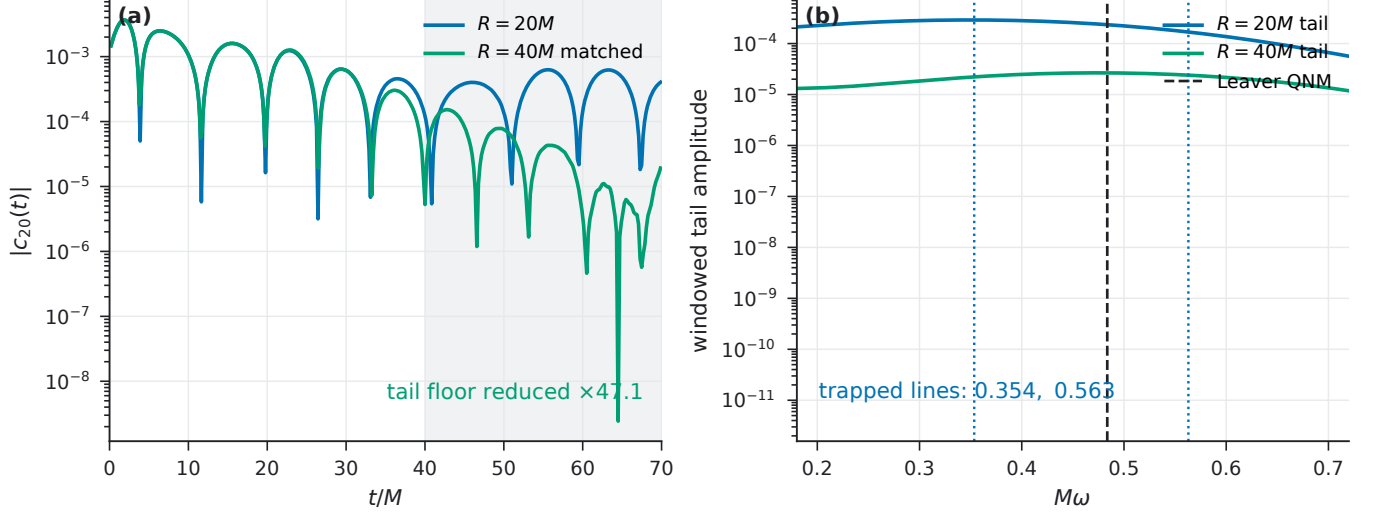


FIG. 5. Paired-domain waveform and tail spectrum. Enlarging the matched domain suppresses the finite-domain tail floor, while the small-domain spectrum contains two persistent trapped lines. The marked lines are finite-domain cavity modes, not physical QNMs; the Leaver frequency is shown only as the physical reference.

The three-dimensional scalar is

$$\phi(0, r, \Omega) = v_0 + g(r)F_{lm}(\Omega), \quad F_{lm} = \begin{cases} 1, & l = 0, \\ Y_{lm}^{\text{real}}, & l > 0, \end{cases} \quad (\text{A2})$$

where the real harmonics are orthonormal and use the same implementation as the multipolar extractor. For the curved ingoing prescription let $\bar{r} = \max(r, w/10)$ and $c_{\text{in}} = \beta^r + \alpha\sqrt{\gamma^{rr}}$. The exact three-dimensional momentum profile is

$$\Pi(0, r, \Omega) = \left[\sqrt{\gamma^{rr}(\bar{r})} g'(r) + \frac{c_{\text{in}}(\bar{r})}{\alpha(\bar{r})} \frac{g(r)}{\bar{r}} \right] F_{lm}(\Omega). \quad (\text{A3})$$

For Schwarzschild–Kerr–Schild, $\sqrt{\gamma^{rr}} = \alpha = (1 + 2M/r)^{-1/2}$, $\beta^r = 2M/(r + 2M)$, and $c_{\text{in}} = 1$. The one-dimensional oracle uses the algebraically equivalent ansatz

$$\Pi_{\text{1D}}(0, r) = \frac{[1 - \beta^r(r)]g'(r) + g(r)/r}{\alpha(r)}. \quad (\text{A4})$$

The production values are $A = 0.10$, $r_0 = 5.0M$, and $w = 1.0M$. The $l = 0$ perturbation deliberately omits a factor Y_{00} , so extraction gives $c_{00} = \sqrt{4\pi}u$; for $l > 0$, $c_{l0} = u_l$. The convention is pinned end to end by the versioned regression tests/test_initial_data_ylm.py.

Appendix B: Numerical protocol

Table II gives the complete interior campaign. Every displayed numerical cell is generated from the canonical number catalog.

The exterior validation protocol is listed in Table III. The small and large domains share the same inner grad-

ing through

$$\ell_c(r) = \ell_{c,\text{in}} + (\ell_{c,\text{out}} - \ell_{c,\text{in}}) \frac{r}{R}, \quad (\text{B1})$$

$$\ell_{c,\text{out}} = \ell_{c,\text{in}} + (\ell_c - \ell_{c,\text{in}}) \frac{R}{20M}, \quad (\text{B2})$$

with $\ell_{c,\text{in}} = \ell_c/3.75$.

TABLE II. Interior production protocol.

Quantity	Discretization or selection	Value
Radial domain	excised ball	$r/M \in [0.1, 15.0]$
Evolution	continuous P_1 , SSP–RK3	CFL = 0.3
Mesh grading	outer $\ell_c = 1.2M$	$\ell_{c,\text{in}}/M \in \{0.056, 0.040, 0.028\}$
Angular banks	$l = 0, 1, 2$	$l_{\text{max}} = 2$ normally; 4 for $l = 2$
Duration	all production pairs	$t_{\text{end}} = 12.0M$
Profile bank	logarithmic spacing, $K = 32$	$r/M \in [0.1, 0.5]$
Initial pulse	curved ingoing Gaussian	$(A, r_0/M, w/M) = (0.10, 5.0, 1.0)$
Observable	primary radial fit and capped strong phase	$r/M \in [0.10, 0.50]$, $t/M \in [4.0, 10.0]$

TABLE III. Exterior spectroscopy protocol.

Quantity	Small domain	Matched large domain
Outer radius	$R = 20M$	$R = 40M$
Damping layer	onset $15M$, width $5M$	onset $30M$, width $10M$
Mesh levels	$\ell_c/M \in \{1.4, 1.0, 0.7\}$	matched levels from Eq. (B2)
Mode and extraction	$(l, m) = (2, 0)$	$r_{\text{ext}} = 6.0M$
Duration	$t_{\text{end}} = 70M$	$t_{\text{end}} = 70M$
Initial pulse	$(A, r_0/M, w/M) = (0.001, 8.0, 2.0)$	identical
Anchored fit	length $26.0M$, search after $12.0M$	Prony modes = 4
Late damping fit	unavailable above the cavity background	length $16.0M$, pooled offset $\geq 10.0M$

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